

Spatial omics data formats

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One experiment, four kinds of data

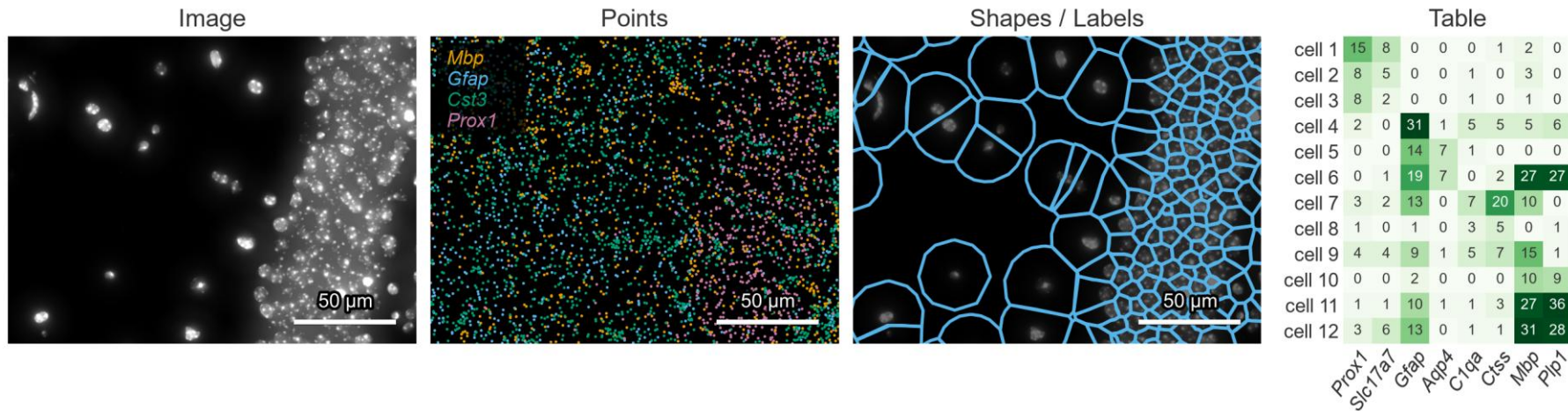
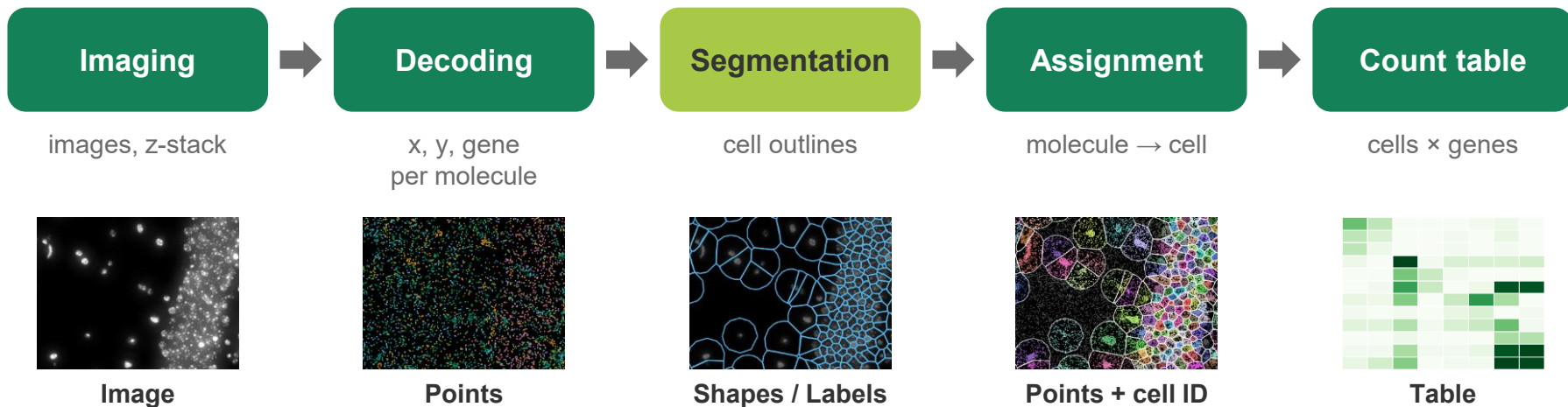


Image: pixels. **Points:** one row per molecule.

Shapes / Labels: one polygon or one label value per cell. **Table:** counts per cell and gene.

From images to a cell × gene table

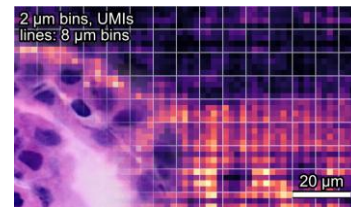
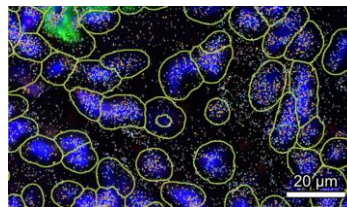
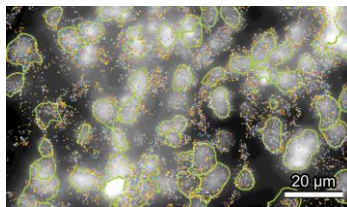
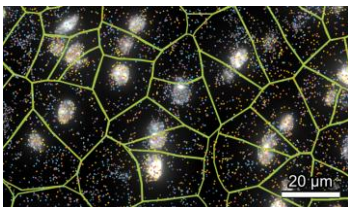
this afternoon



Practical 0 section: 347 genes, about 62,000 cells, about 16 million transcripts, 0.2125 μm per pixel.
About 6 GB of output files, but the count table is only 5 MB.

Every platform, its own file layout

	Xenium	MERSCOPE	CosMx	Visium HD
Counts	cell_feature_matrix.h5	cell_by_gene.csv	exprMat_file.csv	filtered_feature_bc_matrix.h5
Cells	cells.parquet	cell_metadata.csv	metadata_file.csv	tissue_positions.parquet bins
Outlines	cell_boundaries.parquet	cell_boundaries.parquet	CellLabels/*.tif	cell_segmentations.geojson
Molecules	transcripts.parquet μm	detected_transcripts.csv μm	tx_file.csv pixels	<i>none: 2 μm bins</i>
Images	morphology*.ome.tif	images/mosaic_*.tif	CellComposite/*.jpg	spatial/tissue_*.png

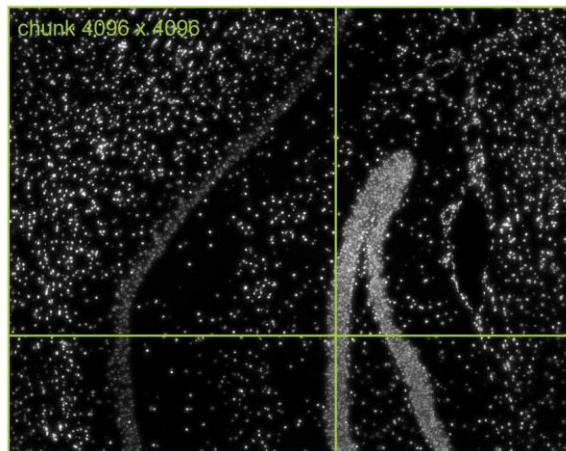


Similar data, different files, names and units.

Five file formats you will meet

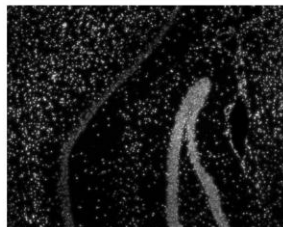
Format	File	What it is	Used for
Parquet	.parquet	A table stored column by column. You can read only the columns you need.	transcripts, cells, polygons
HDF5	.h5 .h5ad	One file. Holds tables, like Parquet, or hierarchical data, like AnnData.	count matrices, AnnData
Zarr	.zarr/	A folder of small chunks plus JSON metadata. Good for very large data and for the cloud.	SpatialData, OME-Zarr
OME-Zarr	.ome.zarr/	Zarr folder for images. OME metadata stored as JSON: channels, pixel size, pyramid levels.	SpatialData images, labels
OME-TIFF	.ome.tif	One TIFF file, with tiles and pyramid levels. Image metadata stored as OME-XML.	vendor images (Xenium)

Chunks and pyramids



s0

7009 x 5559 px
0.2125 $\mu\text{m}/\text{px}$



s1

3505 x 2780 px
0.425 $\mu\text{m}/\text{px}$



s2

1752 x 1390 px
0.85 $\mu\text{m}/\text{px}$



s3

876 x 695 px
1.7 $\mu\text{m}/\text{px}$



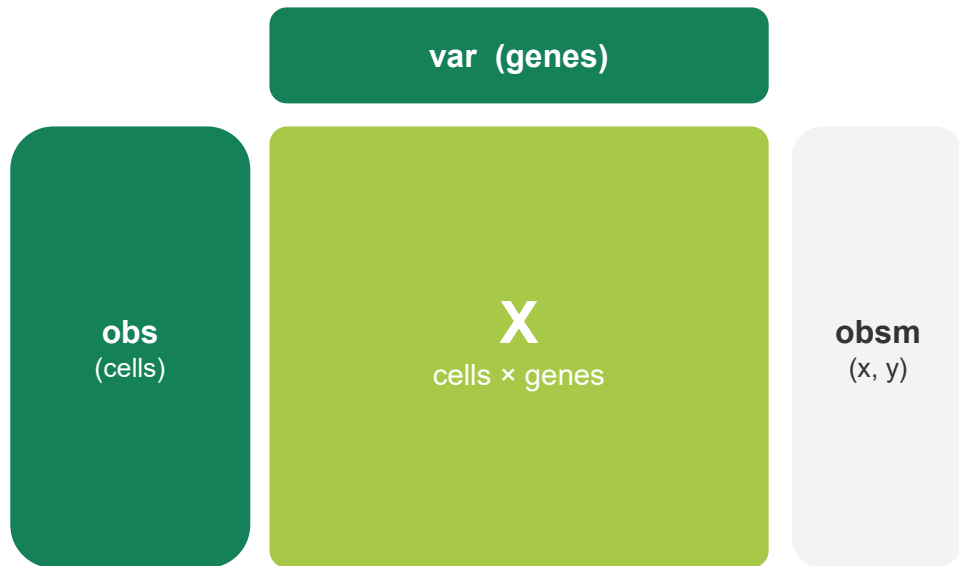
s4

438 x 347 px
3.4 $\mu\text{m}/\text{px}$



Each level is half the size of the one before. A viewer reads only the level and the chunks it needs.

AnnData



Slot	In our data
X	counts, 5,568 cells × 347 genes, sparse
obs	cell_area, total_counts, cell_id
var	gene_ids, feature_types
obsm['spatial']	x, y of each cell
layers	other versions of X, such as normalized counts
uns	everything else



AnnData alone is not enough

- One table per object
- Images hidden in uns, with scale factors to apply by hand
- No place for single molecules or polygons
- H&E, DAPI, transcripts: each in its own pixels. You line them up by hand

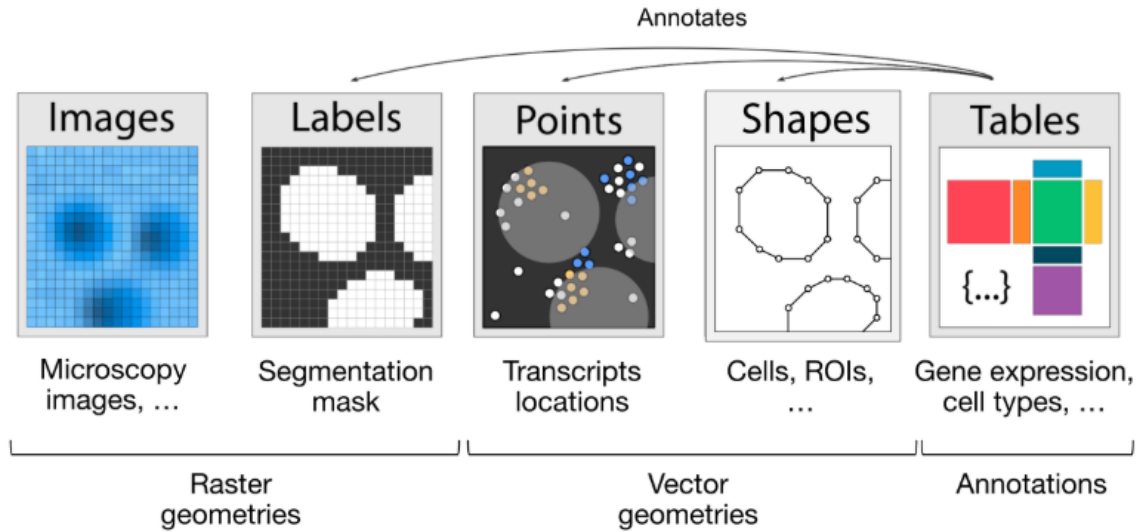
Visium in AnnData

```
adata.obsm["spatial"]           # x, y per spot (pixels)

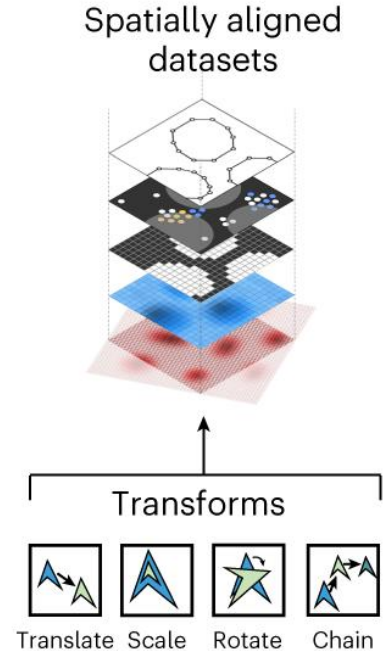
lib = adata.uns["spatial"]["V1_Mouse_Brain_Sagittal_Posterior"]
lib["images"]["hires"]          # the H&E image
lib["scalefactors"]             # full-res pixels -> hires pixels
```



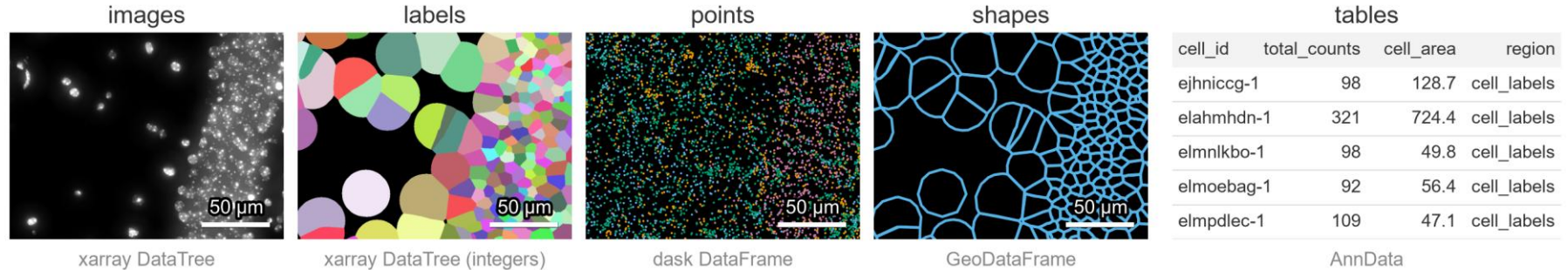
SpatialData



A data model and a Python library (scverse). Stored as Zarr.



The five elements in our data



The table points to the labels: each row is one cell of `cell_labels`. Images, labels and points are read only when needed.

On disk: one Zarr folder

```
Practical1_cropped.zarr/
├── zarr.json
├── images/
│   ├── morphology_focus/          s0 ... s4
│   └── morphology_mip/
│       ├── zarr.json
│       ├── s0/  1 x 5559 x 7009  uint16, chunks 4096
│       ├── s1/  1 x 2780 x 3505
│       ├── s2/  1 x 1390 x 1752
│       ├── s3/  1 x 695 x 876
│       └── s4/  1 x 347 x 438
├── labels/
│   ├── cell_labels/              s0 ... s4 (uint32)
│   └── nucleus_labels/          s0 ... s4
├── points/
│   └── transcripts/points.parquet/ part.0 ... part.3.parquet
├── shapes/
│   ├── cell_boundaries/shapes.parquet
│   └── nucleus_boundaries/shapes.parquet
└── tables/
    └── table/
        ├── X/                   data, indices, indptr (sparse counts)
        └── obs/ var/ obsm/ uns/ layers/ ...
```

images, labels

OME-Zarr pyramids

points

Parquet

shapes

GeoParquet

tables

AnnData in Zarr

Reading data in Python

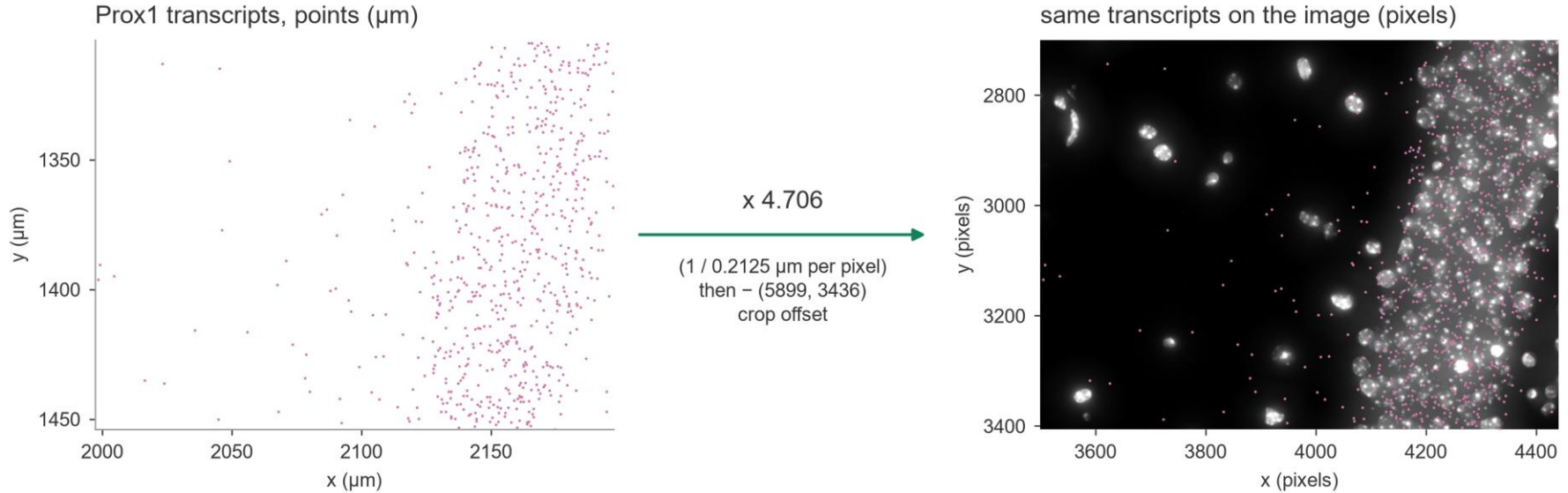
```
from spatialdata_io import xenium
import spatialdata as sd

sdata = xenium("Xenium_V1_FFPE_TgCRND8_17_9_months_outs")    # raw vendor files
sdata.write("xenium.zarr")                                    # once
sdata = sd.read_zarr("xenium.zarr")                           # fast, next time
```

SpatialData object, with associated Zarr store: .../Practical11_cropped.zarr

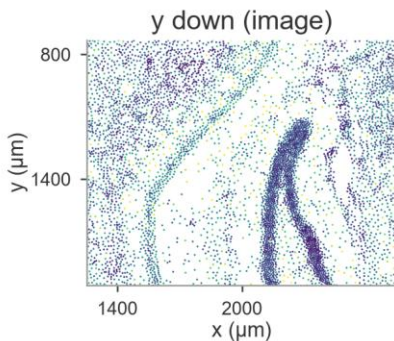
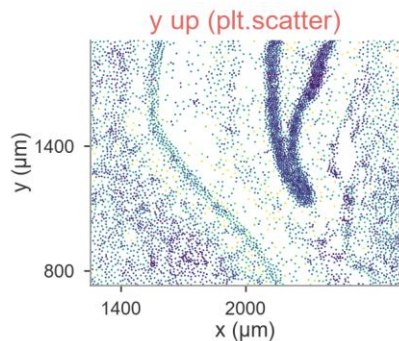
```
├── Images
│   ├── 'morphology_focus': DataTree[cyx] (1, 5559, 7009), (1, 2780, 3505), (1, 1390, 1752), (1, 695, 876), (1, 347, 438)
│   └── 'morphology_mip': DataTree[cyx] (1, 5559, 7009), (1, 2780, 3505), (1, 1390, 1752), (1, 695, 876), (1, 347, 438)
├── Labels
│   ├── 'cell_labels': DataTree[yx] (5559, 7009), (2780, 3505), (1390, 1752), (695, 876), (347, 438)
│   └── 'nucleus_labels': DataTree[yx] (5559, 7009), (2780, 3505), (1390, 1752), (695, 876), (347, 438)
├── Points
│   └── 'transcripts': DataFrame with shape: (1261748, 10) (3D points)
├── Shapes
│   ├── 'cell_boundaries': GeoDataFrame shape: (5569, 1) (2D shapes)
│   └── 'nucleus_boundaries': GeoDataFrame shape: (5422, 1) (2D shapes)
└── Tables
    └── 'table': AnnData (5568, 347)
```

Coordinate systems



In Xenium data, `global` is in **pixels of the morphology image**, not in μm .

Common mistakes



Pixels or μm ?

Check the transformation before you compare coordinates.

Upside-down tissue

Images have y pointing down. Plain scatter plots have y up.

Loading everything

Do not call `.compute()` on 16 million transcripts. Filter first.

Wrong pyramid level

Use a small level to look, the full level to measure.

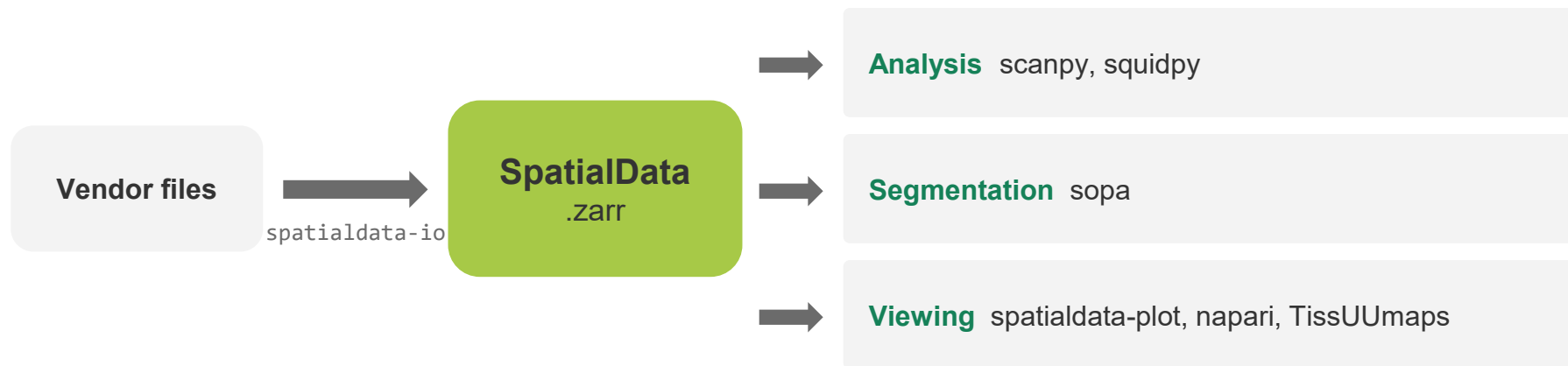
Cell IDs as bytes

Decode them: `.str.decode("utf-8")`

Duplicated gene names

`adata.var_names_make_unique()`

Python and R tools



R SpatialExperiment, SpatialFeatureExperiment (Practical 7), Seurat, Giotto

Python ↔ R .h5ad files: anndataR, zellkonverter

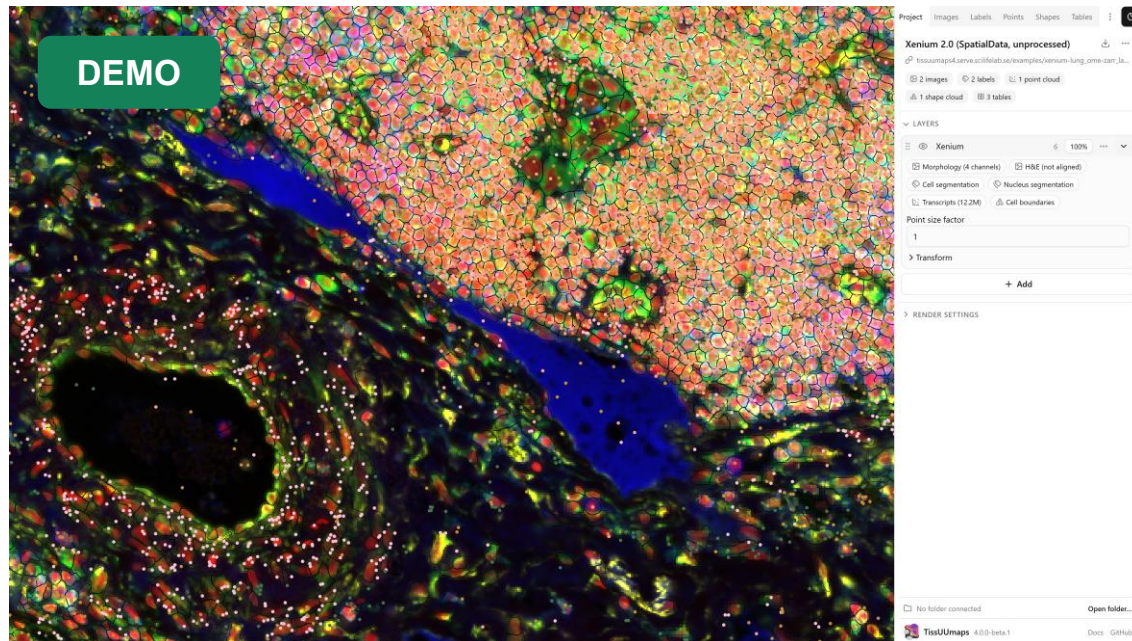
TissUUmeps 4

A web app to view, annotate and share spatial biology data

- Runs in the browser, no server
- Easy to use, like a plotting tool
- Fast: GPU rendering, data loaded on demand
- Images, labels, points, shapes, tables, transforms
- Your own data: local files or cloud formats (OME-Zarr, TIFF, Parquet, AnnData, SpatialData)
- Open source, with plugins

Try it: go.nbis.se/tissuumaps

Code: github.com/TissUUmeps/TissUUmeps



Practical 0

- 1 Build an AnnData by hand from the Xenium files
- 2 Read the same folder with spatialdata-io
- 3 Look at images, labels, points and shapes
- 4 Cut a region with a bounding-box query and save it as Zarr
- 5 Read a Visium HD dataset (8 μm bins)
- 6 **Bonus:** view your saved Zarr in TissUMaps

Notebook

Practical0_DataFormats.ipynb

Keep your output

The files you save are used in the next practicals.

Bonus: TissUMaps

go.nbis.se/tissuumaps

Open a Workspace, then load your local Zarr folder.